

Title 

ACCUMIN: A Monetary Framework for Ending Global Hunger

Author: Vahid Nezamivand Chegane

Email: vahidmaykoltarh@yahoo.com

ORCID: orcid.org/0009-0003-6477-091X

Abstract 

Despite producing enough food for everyone, 318 million people suffer from acute hunger today. This is not a problem of scarcity, but of monetary distribution. When governments print money to help the poor, that money flows through the economy and inflates the price of assets—housing, factories, gold—making the rich richer and leaving the poor behind. This paper introduces ACCUMIN, a two-layer monetary system that separates the economy of survival from the economy of accumulation. In the survival layer, central banks issue non-transferable digital credits for essential goods only (food, medicine, basic housing). These credits cannot be hoarded, cannot be speculated on, and expire monthly. In the accumulation layer, traditional money continues for non-essential goods and investment. The result: zero hunger, zero inflation in essentials, and zero rent-seeking on basic needs—all at a cost of \$47.7 billion per year, which is less than 0.3% of global alcohol spending

Keywords 

Hunger Eradication; Monetary Reform; Basic Income; Inflation Control; CBDC; Poverty Reduction; Rent-Seeking; Speculation

Introduction: The Hunger Paradox .1 

:The world today produces enough food to feed 10 billion people . Yet

Indicator Number Source

Acutely hungry people (2026) 318 million WFP

Children under 5 malnourished 38 million WFP

Countries with famine 7 WFP

Countries with severe hunger 60+ WFP

Total undernourished 800 million+ World Bank

.This is not a resource problem. It is a distribution problem

Why Money Fails the Poor .2 🏠

:When a government prints money to help the poor, it follows a predictable path

- .1 The money enters the economy through government spending or subsidies
- .2 (It flows to businesses, workers, and eventually to asset markets (real estate, stocks, gold)
- .3 Prices of assets rise. Those who already own assets get richer
- .4 The poor, who own no assets, see only the inflation in food and rent

The Inflation Trap .2.1

:Global inflation has been eroding purchasing power for years

Year Global Inflation

8.0% 2022

6.8% 2023

5.9% 2024

4.1% 2025

est.) 3.8%) 2026

est.) 3.4%) 2027

Each percentage point of inflation is a tax on the poor, who spend most of their income on essentials

The Cost of Physical Money .2.2

:Maintaining a cash-based system is expensive

Printing and minting costs millions per country annually ·

Security and transport add billions globally ·

of cash sits outside the banking system, ready for hoarding and speculation 94.2% ·

Crime and Desperation .2.3

Research confirms that poverty, unemployment, and underemployment are direct drivers of crime . When survival is at stake, dignity disappears

ACCUMIN: A Two-Layer Monetary System .3 🧠

ACCUMIN proposes a fundamental redesign of money. Instead of one type of money for everything, we create two separate layers

Layer 1: The Survival Economy .3.1

:This layer handles only essential goods

(Food (bread, milk, rice, chicken, eggs, oil, sugar ·
Basic medicine ·
(Basic housing (rent or mortgage for low-income households ·

:How it works

.Every citizen receives a monthly digital credit from the central bank ·
.This credit can only be spent on approved essentials ·
.The credit expires monthly – no hoarding, no saving, no speculation ·
.Merchants receive credits, which they use to pay suppliers or taxes ·

:Mathematical representation

$$(C_i(t) = B - \sum_{j=1}^n P_j \cdot Q_{ij}(t)$$

:Where

$C_i(t)$ = remaining credit for person i at time t ·
(B = monthly basic credit (fixed ·
 P_j = price of essential good j ·
 Q_{ij} = quantity purchased ·

Layer 2: The Accumulation Economy .3.2

:This layer handles everything else

(Non-essential goods (clothing, electronics, travel ·
(Investment (stocks, real estate, business ·
International trade ·

:How it works

.Traditional money continues to exist here ·
.It can be earned, saved, invested, and speculated on ·
.But it cannot buy essentials – that is reserved for Layer 1 credits ·

Why ACCUMIN Works .4

Zero Hunger .4.1

:The cost to feed all 318 million hungry people is \$47.7 billion per year

times 150 = 47,700,000,000\ 318,000,000

This is less than 0.3% of global alcohol spending (\$2.2 trillion annually). The money exists.
.ACCUMIN provides the mechanism

Zero Inflation in Essentials .4.2

Because Layer 1 credits are fixed in supply and non-transferable, prices of essentials cannot
:rise. The basic credit equation ensures

$$\frac{dP_j}{dt} = 0 \quad \text{for all essential goods } j$$

Zero Rent-Seeking .4.3

Speculators profit by buying essentials cheap and selling them dear, or by hoarding and
:creating artificial scarcity. In ACCUMIN, this is impossible because

- .Credits expire monthly – hoarding is pointless ·
- .Credits are non-transferable – they cannot be sold or exchanged ·

(Comparison with Universal Basic Income (UBI) .4.4

Feature UBI ACCUMIN

Form Cash Non-transferable credit

Use Any purpose Essentials only

Inflation risk High Zero

Hoarding possible Yes No

Speculation possible Yes No

Annual cost (US example) \$1.8 trillion \$48 billion

Feasibility and Technology .5

:ACCUMIN is not a distant dream. All necessary technology exists today

CBDC (Central Bank Digital Currency): Over 100 countries are developing or testing ·
.CBDCs

.Blockchain: Provides transparent, immutable ledgers ·

.National ID systems: Over 70% of the world's population now has a digital ID ·

.Mobile payments: Even in poor regions, mobile money is widespread ·

Projected Impacts .6

Within One Year .6.1

million people fed 318 

Zero hunger-related deaths 

Zero inflation in essential goods ✓
billion annual cost – already available in existing budgets \$47.7 ✓

Within Five Years .6.2

Crime rates drop as survival motive disappears ✓
Economic productivity rises as people focus on growth, not survival ✓
Global inequality decreases for the first time in decades ✓
Trust in institutions restored ✓

Conclusion: A Call to Action .7 

The world has spent trillions on wars, bailouts, and financial speculation. Ending hunger
.costs \$48 billion per year – less than what the world spends on alcohol in two weeks

The problem is not money. The problem is design. ACCUMIN offers a simple, practical, and
technologically feasible redesign of the monetary system to ensure that no human being
.ever goes hungry again

.The framework is ready. The technology exists. The cost is trivial
.What is missing is the decision to act

References 

World Food Programme, Global Outlook 2026 .1
World Bank, Poverty and Shared Prosperity Report 2025 .2
IMF, World Economic Outlook Update, January 2026 .3
(Stanford University, Ending Extreme Poverty: A Costing Study (2025 .4
Central Bank of Trinidad and Tobago, Annual Report 2025 .5
(University of the Philippines, Poverty and Crime: A Statistical Analysis (2025 .6



Author Profiles & Credentials

ORCID (Publications): orcid.org/0009-0003-6477-091X
LinkedIn: linkedin.com/in/vahid-nezamevand-5594593a2
GitHub (Portfolio):
github.com/Vahid-nezamivand-chegane/Repository-name-Technical-Writing-Portfolio
Zenodo Papers: zenodo.org/search?q=nezamivand
Full Certifications (130+): tinyurl.com/Vahid-nezamivand-chegane
Email: vahidmaykoltarh@yahoo.com

